Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python orchestration systems**, **model evaluation workflows**, concurrent services, **observable data pipelines**, and reproducible stateful production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured model evaluation**, model switching, fallback paths, persistent results, and instructor review controls supporting reproducible verification reliably at scale.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized downstream datasets while reducing storage costs by 15% reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, generating failure reports, coverage summaries, and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash
Backend & Concurrency: FastAPI, Flask, REST APIs, Microservices, Multithreading, Async Processing, Background Workers, WebSockets
Data & Orchestration: Data Pipelines, ETL, AWS Glue, AWS S3, PostgreSQL, AWS RDS, Redis, Data Validation, Job State Management
AI & Evaluation: LLMs, Transformers, PyTorch, Model Evaluation, Offline RL, RAG, FAISS, Pinecone
Infrastructure & Reliability: Linux, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Unit Testing, Integration Testing, PyTest, Observability, Error Handling, Debugging

Projects

Agent Evaluation Orchestrator - Design Concept
- Designed an **agent-evaluation orchestrator** concept executing multi-turn scenarios with versioned datasets, retries, isolated state, validators, scoring, trace collection, run comparisons, and reliability metrics across concurrent evaluation workloads.

Stateful Enterprise Simulator - Design Concept
- Designed a **stateful enterprise simulator** concept with persistent ITSM and HR fakes, per-run data injection, deterministic state machines, LLM user personas, contract tests, and hermetic setup and teardown.

Big-Brain | Live
- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.